

Implement ACLs:

- o ACL stands for Access Control List, also called it Access List.
- o ACLs are always processed from top to down in sequential order.
- o A packet is compared with ACL conditions until it finds a match.
- o Once a match is found for packet, no further comparison will be done.
- o In Cisco ASA Firewall Interface will take action based on match condition.
- o In Cisco ASA Firewall ,there are two possible actions permit and deny.
- o If permit condition match, packet will be allowed to pass from interface.
- o If deny condition match, in ASA packet will be destroyed immediately.
- o In Cisco ASA Firewall, every ACL has a default deny statement at end of it.
- o If a packet does not meet with any condition, it will be destroyed by default deny.
- o Implicit deny condition work only if ACL has at least one user defined condition.
- o ACL (Access Control List) can filter only the traffic passing from interfaces.
- o It cannot filter the traffic originated from router on which it has been applied.
- o Only one ACL applied to an interface in each direction inbound and outbound.
- o Only have a single ACL per direction, it is impossible to have 2 inbound access lists.
- o Access Control List work on the network (Layer 3) and the transport (Layer 4) layer.
- o First, create an Access-List globally and then assign it to an Cisco Firewall interface.
- o ACL use for security, deny or permit , filter traffic, accept or reject, allow or deny.
- o The first line of defense in a network is the Access Control List on the edge firewall.
- o Cisco uses Access Control List for many other purposes besides controlling access.
- o ACLs can be used & define which routes will be distributed over a routing protocol.
- o ACL can be used to control Quality of Service (QoS) rules and other policies as well.
- o Access Control List is one of the main features of Cisco Adaptive Security Appliance.
- o Access Control List is used to permit or deny traffic from a source to a destination.



Advantages of ACL:

- o Access Control List, Limits network traffic to increase network performance.
- o ACLs provide traffic flow control by restricting the delivery of routing updates.
- o ACLs controls, which type of traffic, are forwarded or blocked by the router.
- o ACLs helps to increase data security and confidentiality in an organization.
- o Access Control List monitor unauthorized access to important files Access rights.

Cisco ASA ACL Types:

- o Cisco ASA firewall uses name Standard ACL and name Extended ACL.
- o Cisco ASA firewall uses netmask instead of wildcard mask in ACL.
- o Each ACL has a name or numeric ID, such as outside_in, OUTSIDE_IN, or 101.
- o ASA ACL does not use number ranges, but if want can use numbers as a name.
- o ACL is made up of one or more access control entries with same ACL ID or name.
- o ACL's can be applied inbound, outbound and globally for all inbound interfaces.

Extended ACLs:

- o Extended ACLs are the main type that can be used in Cisco ASA Firewall.
- o These ACLs are used for access rules to permit & deny traffic through device.
- o These Access Control List are used for traffic matching by many features.
- o Including service policies, AAA rules, Botnet Traffic Filter, & VPN group policies.
- o Extended ACLs are also supported in routed and transparent Firewall modes.
- o Extended ACLs and Webtype ACLs allow a mix of IPv4 and IPv6 addresses.
- o Access-List ACL_IN extended permit ip any any

EtherType ACLs:

- o EtherType ACLs apply to non-IP Layer-2 traffic on bridge group member interfaces only.
- o Use these rules to permit or drop traffic based on the EtherType value in Layer-2 packet.
- o With EtherType ACLs, you can control the flow of non-IP traffic across the Cisco device.
- o EtherType Access Control Lists are supported in Cisco ASA Firewall transparent mode only.
- o EtherType Access Control Lists (ACLs) do not contain Internet Protocol (IP) addresses.
- o Access List ETHER ethertype deny mpls-multicast.

Webtype ACLs:

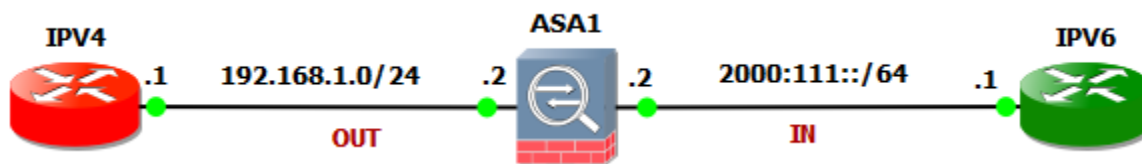
- o The Webtype ACLs are used for filtering clientless SSL VPN traffic.
- o These ACLs can deny access based on URLs or destination addresses.
- o Webtype ACLs are supported in Cisco ASA Firewall routed mode only.
- o Access List accompany Webtype deny url http://*.cisco.com

Standard ACLs:

- o Standard ACLs identify traffic in Cisco ASA by source address only.
- o There are few features that use them: route maps and VPN filters.
- o VPN filters also allow extended ACL, limit standard ACL use to route maps.
- o Standard ACLs are supported in routed and transparent firewall modes.
- o Standard Access Control Lists (ACLs) do not allow IP version 6 addresses.
- o Access List OSPF standard permit 192.168.1.0 255.255.255.0

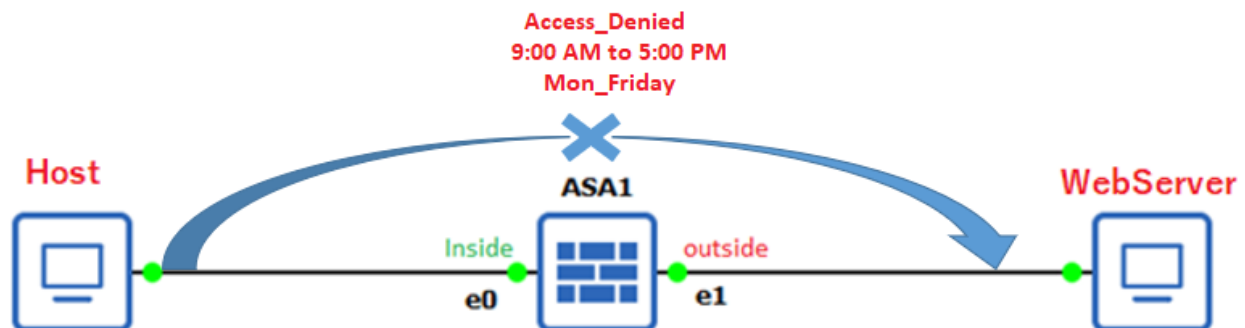
Unified IPv6/IPv4 ACL:

- o ACLs now support Internet Protocol v4 and Internet Protocol v6 address.
- o Prior to version 9.0(1), IPv6 and IPV4 Access Lists was manage separately.
- o Later versions have unified functionality & removed IPv6 specific commands.
- o Single ACL can now include a mixture of both IPv4 & IPV6 hosts and networks.
- o The Object groups can also be include a mixture of these addresses as well.
- o Any keyword has been modified to include both IPv4 and IPv6 addressing.
- o The new 'any4' and 'any6' keywords now refer to IPv4 and IPv6 addresses.
- o Communication will only work if NAT46 and NAT64 has been configured.
- o Also, map the addresses to a valid address in other Internet Protocol version.
- o Without the NAT46 and NAT64, the commands are still valid but will not work.
- o **Access-list TEST extended permit ip 2001:111:1::/64 10.2.2.0 255.0.0.0**



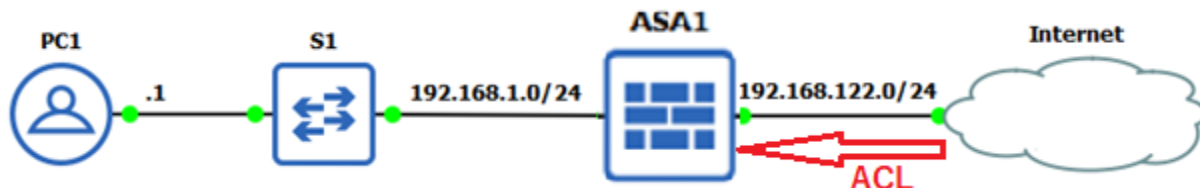
Time-Based ACL:

- o Time-based ACLs that allow for network access based on time or day.
- o The Time-based ACLs are only active during a specified time range.
- o For Time-Based ACLs set time range to be either periodic or absolute.
- o Configure time-based ACL, specify time range and then apply the ACL.
- o To verify, the ACL run show access-lists and check if the ACL is active.
- o If it is in the time range and the Access Control List is filtering traffic.
- o Time-based ACLs allow for access control based on time and date.
- o It uses time ranges to block the traffic during specified duration.
- o Periodic specifies recurring time period for which the time range is valid.
- o Absolute specifies a single time period for which the time range is valid.



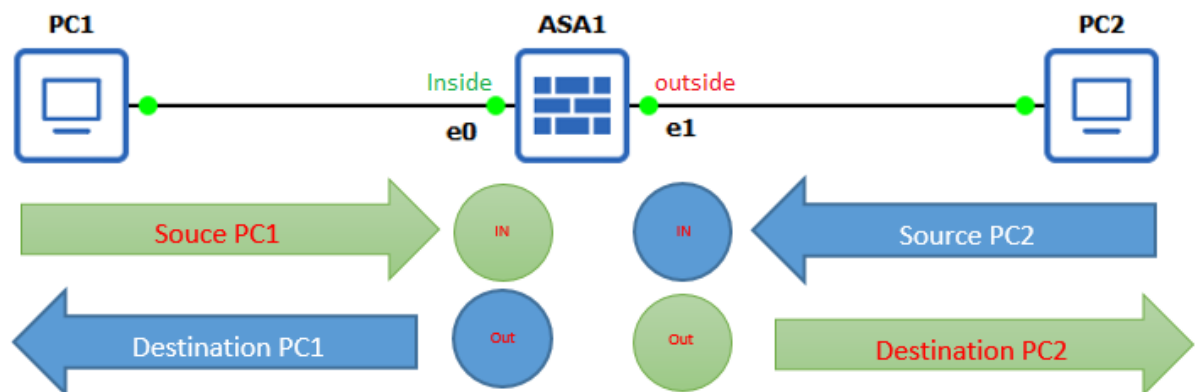
Infrastructure ACL (iACLs):

- o Using the Infrastructure iACLs is a technique that was developed by ISPs.
- o These iACLs are now a common practice by enterprises and other organizations.
- o iACLs involves use of ACLs that prevent direct attacks to infrastructure devices.
- o Configure these ACLs to specifically allow only authorized traffic to infrastructure.
- o ICMP Packet Filtering, special addresses, Private addresses and own public range.



ACL Direction:

An ACL applied inbound affects traffic entering the ASA. An ACL applied outbound affects traffic leaving the ASA. So, a connection initiated from the outside to an inside host will arrive at the ASA on the outside interface and is entering the ASA so the inbound ACL on the outside interface. Act like you are standing 'in the interface' and face in the direction the interface is pointing, away from the ASA. Then think about incoming and outgoing.



Global Access List:

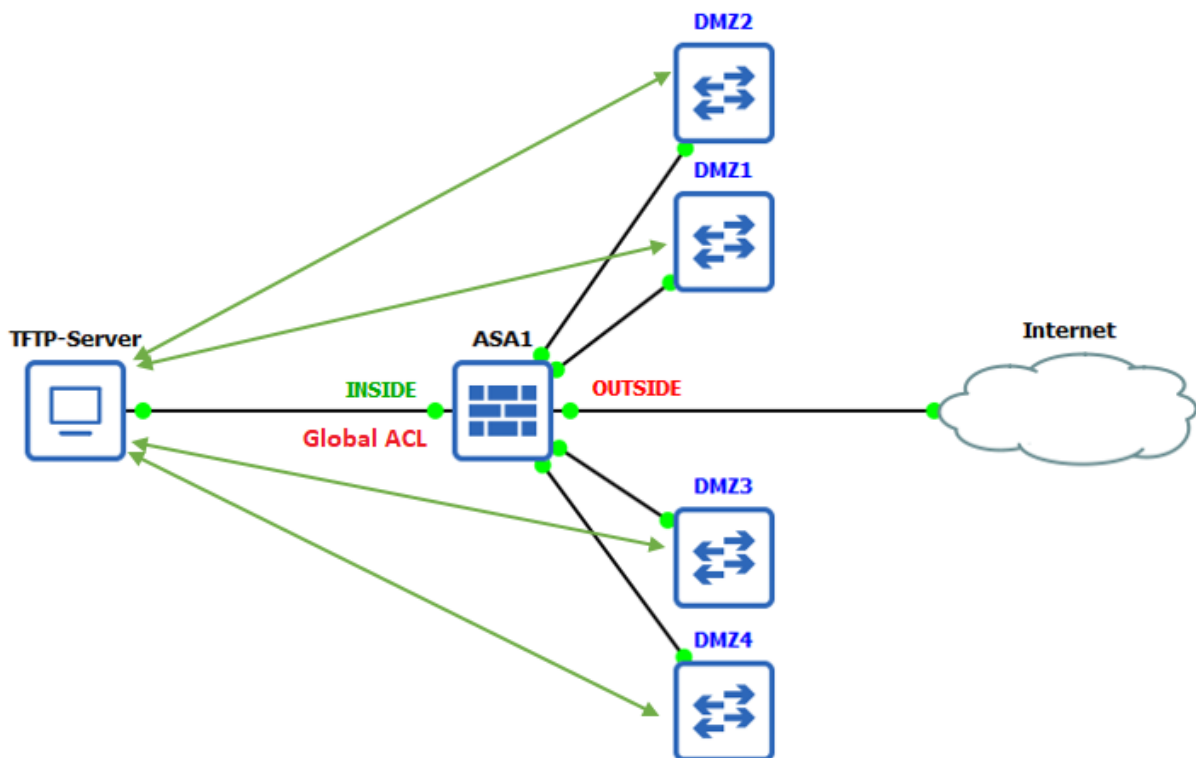
Global access list applies logically to the entire firewall in inbound direction to all interface. If there are existing interface access lists, those will be considered first and instead of having implicit deny any any at the end of interface ACLs, the Global access list is processed and in case of non-matching rule, the implicit deny any any is used at the end of Global access list. This is an access list that will allow traffic inbound on all interfaces. To create global access list using ASDM open access rule, add access rule, and for interface choose Any.

Policy Ordering

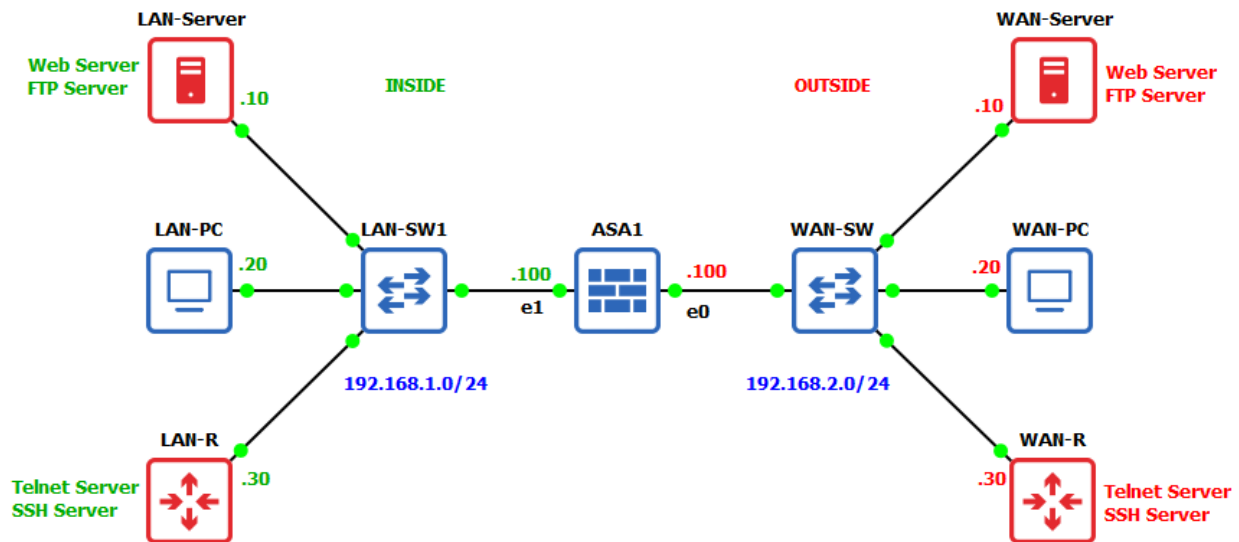
Interface Specific access-list

Global access-list

Default (implicit) deny ip any any



Lab Time:



LAN-Server Toolbox Configuration	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.10 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 192.168.1.100 > /etc/resolv.conf
LAN-PC Webterm Configuration	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.20 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 192.168.1.100 > /etc/resolv.conf
LAN-R Configuration	LAN-R(config)#interface e0/0 LAN-R(config-if)#ip add 192.168.1.30 255.255.255.0 LAN-R(config-if)#no shutdown LAN-R(config-if)#exit LAN-R(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.100
WAN-R Configuration	WAN-R(config)#interface e0/0 WAN-R(config-if)#ip add 192.168.2.30 255.255.255.0 WAN-R(config-if)#no shutdown WAN-R(config-if)#exit WAN-R(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.100

WAN-Server Toolbox Configuration	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.2.10 netmask 255.255.255.0 gateway 192.168.2.100 up echo nameserver 192.168.2.100 > /etc/resolv.conf
WAN-PC Webterm Configuration	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.2.20 netmask 255.255.255.0 gateway 192.168.2.100 up echo nameserver 192.168.2.100 > /etc/resolv.conf
Cisco ASA1 Configuration	ASA1(config)# interface e0 ASA1(config-if)# ip add 192.168.2.100 ASA1(config-if)# nameif outside ASA1(config-if)# no shutdown ASA1(config-if)# exit ASA1(config)# interface e1 ASA1(config-if)# ip add 192.168.1.100 ASA1(config-if)# nameif inside ASA1(config-if)# no shutdown ASA1(config)# fixup protocol icmp

Telnet and SSH Configuration	
LAN-R Telnet and SSH Server Configuration	LAN-R(config)#username admin password 123 LAN-R(config)#ip domain-name test LAN-R(config)#crypto key generate rsa LAN-R(config)#line vty 0 4 LAN-R(config-line)#transport input all LAN-R(config-line)#login local LAN-R(config)#enable password 123
WAN-R Telnet and SSH Server Configuration	WAN-R(config)#username admin password 123 WAN-R(config)#ip domain-name test WAN-R(config)#crypto key generate rsa WAN-R(config)#line vty 0 4 WAN-R(config-line)#transport input all WAN-R(config-line)#login local WAN-R(config)#enable password 123

- o Check and Verify Deny traffic from Inside to Outside zone in Cisco ASA Firewall.
- o Check and Verify Permit Traffic from outside to Inside zone in Cisco ASA Firewall.
- o Check and Verify Editing Access Control Lists (ACL) in Cisco ASA Firewall.
- o Check and Verify Rename Access Control Lists (ACL) in Cisco ASA Firewall.
- o Check and Verify Global Access Control List (ACL) in Cisco ASA Firewall.
- o Check and Verify Remove Access Control List (ACL) in Cisco ASA Firewall.

Permit Traffic Outside to Inside
ASA1(config)#access-list out2in permit tcp host 192.168.2.20 host 192.168.1.10 eq 80
ASA1(config)# access-group out2in in interface outside
ASA1(config)# show access-list
ASA1# show run access-list
ASA1# show run access-group
ASA1(config)# clear configure access-list in2out
ASA1(config)# clear configure access-group

Deny Traffic Inside to Outside
ASA1(config)# access-list in2out deny tcp host 192.168.1.20 host 192.168.2.10 eq www
ASA1(config)# access-list in2out extended permit ip any any
ASA1(config)# access-group in2out in interface inside
ASA1(config)# access-list in2out rename in2web
ASA1(config)# show access-list
ASA1(config)# show running-config access-group

Infrastructure ACL
!--ICMP Packet Filtering
ASA1(config)#access-list INF extended permit icmp any any echo-reply
ASA1(config)#access-list INF extended permit icmp any any unreachable
ASA1(config)#access-list INF extended permit icmp any any
ASA1(config)#access-list INF extended permit icmp any any time-exceeded
ASA1(config)#access-list INF extended deny icmp any any
!--Deny Special-Use Address Sources
ASA1(config)#access-list INF extended deny ip host 0.0.0.0 any
ASA1(config)#access-list INF extended deny ip 127.0.0.0 255.0.0.0 any
ASA1(config)#access-list INF extended deny ip 192.0.2.0 255.255.255.0 any
ASA1(config)#access-list INF extended deny ip host 224.0.0.0 any
!--Filter Private Space.
ASA1(config)#access-list INF extended deny ip 10.0.0.0 255.0.0.0 any
ASA1(config)#access-list INF extended deny ip 172.16.0.0 255.240.0.0 any
ASA1(config)#access-list INF extended deny ip 192.168.0.0 255.255.255.0 any
ASA1(config)# access-list INF extended permit ip any any
ASA1(config)#access-group INF in interface outside

Time-Based ACL

```
ASA1(config)# time-range office
```

```
ASA1(config-time-range)# periodic weekdays 9:00 to 17:00
```

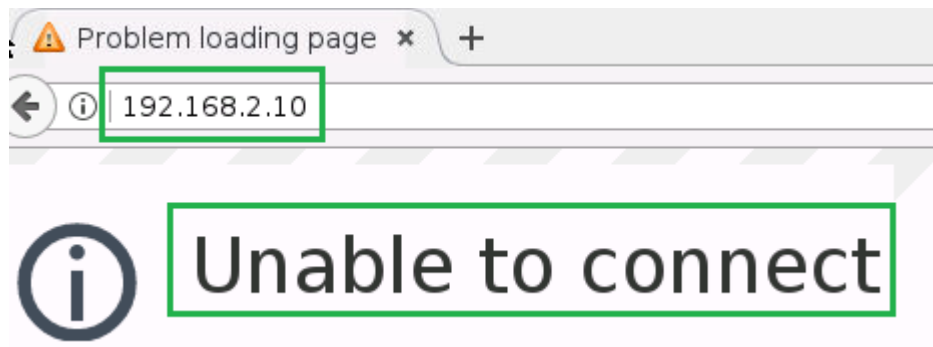
```
ASA1(config-time-range)# exit
```

```
ASA1(config)# access-list webblock extended deny tcp any host 192.168.2.10 eq www time-range office
```

```
ASA1(config)# access-list webblock extended permit ip any any
```

```
ASA1(config)# access-group webblock in interface inside
```

```
ASA1(config)# show access-list webblock
access-list webblock; 2 elements; name hash: 0x19725476
access-list webblock line 1 extended deny tcp any host 192.168.2.10 eq www time-range office (hitcnt=24)
access-list webblock line 2 extended permit ip any any (hitcnt=7) 0x2be8330b
ASA1(config)#
```



```
LXTerminal
File Edit Tabs Help
root@LAN-PC:~#
root@LAN-PC:~# ping 192.168.2.10
PING 192.168.2.10 (192.168.2.10) 56(84) bytes of data.
64 bytes from 192.168.2.10: icmp_seq=1 ttl=64 time=39.0 ms
64 bytes from 192.168.2.10: icmp_seq=2 ttl=64 time=10.3 ms
^C
--- 192.168.2.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/mdev = 10.336/24.687/39.039/14.352 ms
root@LAN-PC:~#
```